# Study Protocol: GENIE BPC Breast Cancer Feature Record

| Field | Value |
|---|---|
| Protocol version | 0.1.0 (draft, status: proposed) |
| Date | 2026-10-09 |
| Format | DICOM PS3.3 Annex A structure (ADR-0003) |
| Source dictionary | `studies/genie_bpc_brca/dictionary.csv`, 153 rows, 136 variables, 40 absent in cohort |
| Dictionary SHA-256 | `abb7d28476616c74dbb7f40b4d99a948837972992a80f0fdf765d9dd39728c42` |
| Generated by | protocol generator from the dictionary; hand refinements to follow in `protocol.yaml` |

## Key

**Module usage** (PS3.3 A.1.3): **M** mandatory; **C** conditional, required when the condition holds and forbidden otherwise; **U** user option, validated when present.

**Attribute types** (PS3.5 7.4, plus harness type X):

| Type | Meaning | Harness rule |
|---|---|---|
| 1 | Required, non-empty | missing or empty rejects the record |
| 1C | Required, non-empty, if a condition holds | as type 1 when the condition holds |
| 2 | Required, may be empty | key present; `null` allowed with `status = not_recorded` |
| 2C | Required, may be empty, if a condition holds | present exactly when the condition holds |
| 3 | Optional | may be omitted |
| X | Absent in cohort | always `null`, `origin = absent`, never sent to the model |

**Origin**: `deterministic` (dictionary mapping or alias), `deterministic; llm if text source` (model gap-fill allowed only when a free-text Source Series is present), `regex` (drug-class scan), `computed` (derived from other attributes), `absent`.

**Gates**: see P.6.

## P.1 IOD Description

**Real-world object.** One patient's research feature record in the AACR Project GENIE Biopharma Collaborative (BPC) breast cancer cohort: the validated `x_study_features` object plus the evidence, verification, and review trail for every value.

**Scope.**

- De-identified research use only. Not a clinical document; never written to an EHR.
- One record per (patient, study, protocol version).

- **T = 0:** date of the index breast cancer diagnosis. Every time is stored as days or months from T = 0.
- **Sources:** BPC curated registry tables (structured); pathology report text where available (free text, model gap-fill only).
- **Output sections:** `clinical`, `pathology`, `follow_up`, `treatment`.

## P.2 IOD Entity-Relationship Model

```
Patient --1-- is the subject of --1-n-- Study (cohort + protocol version)
Study --1-- contains --1-n-- Source Series (one source type each)
Study --1-- contains --1-- Feature Record (x_study_features)
Temporal Frame of Reference --1-- temporally defines --1-n-- Source Series
Extractor (Equipment) --1-- creates --1-n-- Extraction Run
Source Series --1-- contains --1-n-- Source Instance (row | report | PDF)
Feature Record --1-- contains --1-n-- Feature Value
Feature Value --1-- references --1-n-- Evidence Annotation -> Source Instance
Feature Value --1-- has --0-n-- Verification Result
Feature Record --1-- has --0-1-- Approval
```

**Context rules.**

1. All Source Instances in a Study share one Patient and Study context; a `patient_id` disagreement rejects the record.
2. A Source Series holds one source type from one source system.
3. Every Source Series shares one Temporal Frame of Reference (T = 0 above).

## P.3 IOD Module Table

| IE | Module | Usage | Condition | Output section | Attributes | Type X |
|---|---|---|---|---|---|---|
| Study | General Study | M | | `clinical` | 1 | 0 |
| Patient | Patient | M | | `clinical` | 12 | 1 |
| Feature Record | Clinical | M | | `clinical` | 11 | 3 |
| Feature Record | Pathology | M | | `pathology` | 34 | 12 |
| Feature Record | Sequencing | M | | `pathology` | 22 | 13 |
| Feature Record | Follow-up | M | | `follow_up` | 42 | 11 |
| Feature Record | Competing Risks | C | Required when the study includes the Aim 2 brain-metastasis competing-risks analysis (true for this study). | `follow_up` | 4 | 0 |
| Feature Record | Treatment | M | | `treatment` | 10 | 0 |

Harness modules present in every record (defined in the architecture docs, not dictionary-derived): **PHI Gate** (M), **Verification Result** (M), **Consensus** (C: required if any value is model-filled), **Approval** (C: required if the record was routed to review), **Ledger Common** (M). **Intraoperative** (C) is reserved for surgical studies and not used here.

## P.4 Functional Group Macros

**Field Provenance Macro** - applies to every attribute in P.5.

| Attribute | Type | Meaning |
|---|---|---|
| `value` | 2 | typed value or `null` |
| `status` | 1 | `observed`, `not_recorded`, `ambiguous`, `conflicting` |
| `evidence` | 1C | source field path, or text span with `char_start`, `char_end`, `quote`, `source_sha256`; required when `status = observed` |
| `normalization_note` | 3 | present only when a transformation was applied |
| `origin` | 1 | `deterministic`, `regex`, `computed`, `llm`, `reviewer`, `absent` |
| `extractor_ref` | 1C | required when `origin = llm` |
| `agreement` | 1C | k of n runs agreeing; required when `origin = llm` |
| `gate_results` | 1 | gate IDs evaluated, with pass, fail, or skipped |
| `reviewer` | 1C | pseudonymous reviewer ID and time; required when `origin = reviewer` |

## P.5 Module Attribute Tables

### P.5.1 General Study module (M)

| Attribute | Type | Description | Allowed values / unit | Source mapping | Origin | Gates |
|---|---|---|---|---|---|---|
| `study_id` | 1 | long-form alias for canonical source | as source | `cohort` | deterministic | AL |

### P.5.2 Patient module (M)

| Attribute | Type | Description | Allowed values / unit | Source mapping | Origin | Gates |
|---|---|---|---|---|---|---|
| `race_clean` | 2 | from PRIMARY_RACE | White, Black, Asian, Native American | `PRIMARY_RACE`: White / Middle Eastern → White; Black or African American → Black; Asian → Asian; Native American / Alaska → Native American | deterministic | E |
| `ethnicity_clean` | 2 | from ETHNICITY | Non-Hispanic, Hispanic | `ETHNICITY`: Non-Spanish; Non-Hispanic → Non-Hispanic; Hispanic / Latino variants → Hispanic | deterministic | E |
| `patient_id` | 1 | long-form alias for canonical source | as source | `record_id` | deterministic | AL |
| `center` | 1 | long-form alias for canonical source | as source | `CENTER` | deterministic | AL |

| Attribute | Type | Description | Allowed values / unit | Source mapping | Origin | Gates |
|---|---|---|---|---|---|---|
| sex | 2 | long-form alias for canonical source | as source | SEX | deterministic | AL |
| year_of_birth | 2 | long-form alias for canonical source | year | BIRTH_YEAR | deterministic | AL, G3 |
| primary_race | 2 | long-form alias for canonical source | as source | PRIMARY_RACE | deterministic | AL |
| secondary_race | 2 | long-form alias for canonical source | as source | naaccr_race_code_secondary | deterministic | AL |
| tertiary_race | 2 | long-form alias for canonical source | as source | naaccr_race_code_tertiary | deterministic | AL |
| ethnicity_category | 2 | long-form alias for canonical source | as source | ETHNICITY | deterministic | AL |
| region_of_patient_origin | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| patient_id_in_supp_table_1 | 2 | long-form alias for canonical source | as source | record_id | deterministic | AL |

### P.5.3 Clinical module (M)

| Attribute | Type | Description | Allowed values / unit | Source mapping | Origin | Gates |
|---|---|---|---|---|---|---|
| age_cat | 2 | from age_dx | <50, 50-70, >70 | age_dx : <50 → <50; 50-70 (50 ≤ age ≤ 70) → 50-70; >70 → >70 | deterministic | E, G3 |
| age_at_diagnosis | 2 | long-form alias for canonical source | as source | age_dx_num | deterministic | AL, G3 |
| age_at_initial_diagnosis_years | 2 | long-form alias for canonical source | years | age_dx_num | deterministic | AL, G3 |
| age_at_primary_diagnosis | 2 | long-form alias for canonical source | as source | age_dx_num | deterministic | AL, G3 |
| cancer_type | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| cancer_type_detailed | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| additional_breast_cancer_diagnoses | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| number_of_other_invasive_cancer_diagnosis | 2 | long-form alias for canonical source | as source | n_cancers_index | deterministic | AL |
| other_invasive_cancer_diagnosis | 2 | long-form alias for canonical source | as source | non_idx_types | deterministic | AL |
| number_of_cancers_any_type | 2 | long-form alias for canonical source | as source | n_cancers | deterministic | AL |
| number_of_bpc_project_cancers_index_cancers | 2 | long-form alias for canonical source | as source | n_cancers_index | deterministic | AL |

### P.5.4 Pathology module (M)

| Attribute | Type | Description | Allowed values / unit | Source mapping | Origin | Gates |
|---|---|---|---|---|---|---|
| sample_type_bin | 2 | from SAMPLE_TYPE_DETAILED | 1, 2 | SAMPLE_TYPE_DETAILED: Primary tumor → 1; Metastasis → 2 | deterministic | E |
| grade_ord | 2 | from ca_grade | Low, Intermediate, High | ca_grade: I/Low → Low; II/Intermediate → Intermediate; III/High → High | deterministic; llm if text source | E, AN |
| stage_diag_group | 2 | from stage_dx | Stage I, Stage II, Stage III, Stage IV | stage_dx: I/IA/IB → Stage I; II/IIA/IIB → Stage II; III/IIIA-C → Stage III; IV → Stage IV | deterministic; llm if text source | E, G1, AN |
| stage_iv_bin | 2 | from stage_dx_iv | 1, 0 | stage_dx_iv: Stage IV → 1; Stage I-III → 0 | deterministic | E, G1 |
| receptor_primary_cat | 2 | from bca_subtype | HR+/HER2-, HR+/HER2+, HR-/HER2+, Triple Negative | bca_subtype: HR+, HER2- → HR+/HER2-; HR+, HER2+ → HR+/HER2+; HR-, HER2+ → HR-/HER2+; TNBC → Triple Negative | deterministic; llm if text source | E, G4, AN |
| her2_status_bin | 2 | from ca_bca_her_summ | 1, 0 | ca_bca_her_summ: Positive/elevated/amplified → 1; Negative → 0 | deterministic; llm if text source | E, G4, AN |
| was_a_biopsy_performed_of_metastatic_site | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| oncotree_code | 2 | long-form alias for canonical source | as source | ONCOTREE_CODE | deterministic | AL, O |
| tumor_registry_icd_o_3_behavior_code | 2 | long-form alias for canonical source | as source | naaccr_behavior_cd | deterministic | AL, O |
| other_oncotree_diagnosis | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| other_oncotree_diagnosis_1 | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| other_oncotree_diagnosis_2 | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| other_oncotree_diagnosis_3 | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| ki67_percent | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| oncotype_dx_score | 2C | long-form alias for canonical source. Required when receptor_primary_cat = HR+/HER2-. | as source | ca_bca_oncotypedx | deterministic; llm if text source | AL, AN, C |

| Attribute | Type | Description | Allowed values / unit | Source mapping | Origin | Gates |
|---|---|---|---|---|---|---|
| ajcc_stage | 2 | long-form alias for canonical source | as source | stage_dx | deterministic; llm if text source | AL, G1, AN |
| stage_at_diagnosis | 2 | long-form alias for canonical source | as source | stage_dx | deterministic | AL, G1 |
| grade | 2 | long-form alias for canonical source | as source | ca_grade | deterministic; llm if text source | AL, AN |
| her2_status | 2 | long-form alias for canonical source | as source | ca_bca_her_summ | deterministic; llm if text source | AL, G4, AN |
| hormone_receptor_status | 2 | long-form alias for canonical source | as source | bca_subtype | deterministic; llm if text source | AL, G4, AN |
| er_pr_receptor_change | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| discordance_receptor_status_with_primary | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| her2_receptor_status_at_metastatic_diagnosis | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| her2_receptor_status_at_initial_diagnosis | 2 | long-form alias for canonical source | as source | ca_bca_her_summ | deterministic | AL, G4 |
| hr_status_at_metastatic_diagnosis | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| hr_status_at_initial_diagnosis | 2 | long-form alias for canonical source | as source | bca_subtype | deterministic | AL, G4 |
| site_of_sample_tested | 2 | long-form alias for canonical source | as source | SAMPLE_TYPE_DETAILED | deterministic | AL |
| stage_at_initial_breast_cancer_diagnosis | 2 | long-form alias for canonical source | as source | stage_dx | deterministic | AL, G1 |
| subtype_of_metastatic_diagnosis | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| subtype_of_initial_diagnosis | 2 | long-form alias for canonical source | as source | bca_subtype | deterministic | AL, G4 |
| histology | 2 | long-form alias for canonical source | as source | ca_histology | deterministic; llm if text source | AL, O, AN |
| histology_category | 2 | long-form alias for canonical source | as source | ca_hist_brca | deterministic; llm if text source | AL, O, AN |

| Attribute | Type | Description | Allowed values / unit | Source mapping | Origin | Gates |
|---|---|---|---|---|---|---|
| sample_site | 2 | long-form alias for canonical source | as source | SAMPLE_TYPE_DETAILED | deterministic | AL |
| sample_class | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |

### P.5.5 Sequencing module (M)

| Attribute | Type | Description | Allowed values / unit | Source mapping | Origin | Gates |
|---|---|---|---|---|---|---|
| sample_id | 1 | long-form alias for canonical source | as source | SAMPLE_ID | deterministic | AL |
| sequence_assay_id | 2 | long-form alias for canonical source | as source | SEQ_ASSAY_ID | deterministic | AL |
| age_at_which_sequencing_was_reported | 2 | long-form alias for canonical source | as source | AGE_AT_SEQUENCING | deterministic | AL |
| age_at_sample_collection_4 | 2 | long-form alias for canonical source | as source | AGE_AT_SEQUENCING | deterministic | AL |
| age_at_sequencing | 2 | long-form alias for canonical source | as source | AGE_AT_SEQUENCING | deterministic | AL |
| sequenced_sample | 2 | long-form alias for canonical source | as source | SAMPLE_ID | deterministic | AL |
| how_long_sample_was_collected_after_starting_first_line_systemic_therapy_30_days_30_days_treatment_naive | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| number_of_samples_per_patient | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| sequencing_method | 2 | long-form alias for canonical source | as source | SEQ_ASSAY_ID | deterministic | AL |
| fraction_genome_altered | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| mutation_count | 2 | long-form alias for canonical source | as source | mutation_count_all_sites_sum | deterministic | AL |
| mismatch_repair_mmr_testing_at_time_of_sample_acquisition | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| msi_h_test_result_at_time_of_sample_acquisition | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| microsatellite_instability_msi_testing_at_time_of_sample_acquisition | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| tumor_mutational_burden | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |

| Attribute | Type | Description | Allowed values / unit | Source mapping | Origin | Gates |
|---|---|---|---|---|---|---|
| tmb_nonsynonymous | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| x1p_19q_codeletion | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| idh1_2_mutation | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| integrated_histomolecular_group | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| mgmt_methylation_status | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| interval_between_sequencing_and_metastatis_daignosis | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| year_of_next_generation_sequencing | 2 | long-form alias for canonical source | year | CPT_SEQ_DATE | deterministic | AL |

### P.5.6 Follow-up module (M)

| Attribute | Type | Description | Allowed values / unit | Source mapping | Origin | Gates |
|---|---|---|---|---|---|---|
| patients_age_of_death_in_days | 2C | long-form alias for canonical source. Required when vital status = deceased. **Open issue:** Name says days; source age_death_yrs is in years. | days | age_death_yrs | deterministic | AL, C |
| age_at_first_distant_metastasis_in_days | 2C | long-form alias for canonical source. Required when any distant metastasis = 1. **Open issue:** Name says age; source dx_to_dmets_days is days from diagnosis. | days | dx_to_dmets_days | deterministic | AL, C |
| time_from_initial_diagnosis_to_metastatic_diagnosis_months | 2C | long-form alias for canonical source. Required when any distant metastasis = 1. | months | dx_to_dmets_mos | deterministic | AL, G2, C |
| cause_of_death | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| patients_vital_status | 2 | long-form alias for canonical source | as source | os_dx_status | deterministic | AL |
| vital_status | 2 | long-form alias for canonical source | as source | os_status_f | deterministic | AL |
| overall_survival_months | 2 | long-form alias for canonical source | months | OS_months | deterministic | AL, G2 |
| overall_survival_status | 2 | long-form alias for canonical source | as source | os_dx_status | deterministic | AL |

| Attribute | Type | Description | Allowed values / unit | Source mapping | Origin | Gates |
|---|---|---|---|---|---|---|
| patients_age_of_last_follow_up_in_days | 2 | long-form alias for canonical source | days | last_anyvisit_int | deterministic | AL |
| interval_in_days_from_dob_to_date_of_last_contact | 2 | long-form alias for canonical source | days | dob_lastalive_int | deterministic | AL |
| interval_in_days_from_dob_to_dod | 2C | long-form alias for canonical source. Required when vital status = deceased. | days | hybrid_death_int | deterministic | AL, C |
| year_of_last_contact | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| year_of_death | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| pfs_i_from_diagnosis_status | 2 | long-form alias for canonical source | as source | pfs_i_adv_status | deterministic | AL |
| pfs_m_from_diagnosis_status | 2 | long-form alias for canonical source | as source | pfs_m_adv_status | deterministic | AL |
| age_at_metastatic_diagnosis_years | 2C | long-form alias for canonical source. Required when any distant metastasis = 1. | years | dx_to_dmets_yrs | deterministic | AL, C |
| lrr_date_interval_days | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| lrr_radiation_therapy | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| site_of_lrr | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| distant_metastatic_disease_interval | 2C | long-form alias for canonical source. Required when any distant metastasis = 1. | as source | dx_to_dmets_days | deterministic | AL, C |
| distant_mets_adrenal | 2 | long-form alias for canonical source | as source | dist_mets_adrenal | deterministic | AL, G7 |
| distant_mets_bone | 2 | long-form alias for canonical source | as source | dist_mets_bone | deterministic | AL, G7 |
| distant_mets_brain | 2 | long-form alias for canonical source | as source | dist_mets_brain_cns | deterministic | AL, G6 |
| distant_mets_liver | 2 | long-form alias for canonical source | as source | dist_mets_liver | deterministic | AL, G7 |
| distant_mets_lung | 2 | long-form alias for canonical source | as source | dist_mets_pulmonary | deterministic | AL, G7 |
| distant_mets_lymph_nodes | 2 | long-form alias for canonical source | as source | dist_mets_lymph_nodes | deterministic | AL, G7 |
| distant_mets_other | 2 | long-form alias for canonical source | as source | dist_mets_other | deterministic | AL, G7 |
| distant_mets_pleura | 2 | long-form alias for canonical source | as source | dist_mets_pleura_and_malignant_pleural_effusion | deterministic | AL, G7 |

| Attribute | Type | Description | Allowed values / unit | Source mapping | Origin | Gates |
|---|---|---|---|---|---|---|
| distant_mets_subcutaneous_tissue | 2 | long-form alias for canonical source | as source | dist_mets_skin | deterministic | AL, G7 |
| age_at_metastatic_diagnosis | 2C | long-form alias for canonical source. Required when any distant metastasis = 1. **Open issue:** Name says age; source dx_to_dmets_mos is months from diagnosis. | as source | dx_to_dmets_mos | deterministic | AL, C |
| met_site_liver | 2 | long-form alias for canonical source | as source | dist_mets_liver | deterministic | AL |
| has_this_patient_experienced_a_loco_regional_recurrence_lrr | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| met_site_lung | 2 | long-form alias for canonical source | as source | dist_mets_pulmonary | deterministic | AL |
| met_site_lymph_node | 2 | long-form alias for canonical source | as source | dist_mets_lymph_nodes | deterministic | AL |
| met_site_multiple_sites | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| met_site_other_site | 2 | long-form alias for canonical source | as source | dist_mets_other | deterministic | AL |
| met_site_bone_only | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| met_site_bone | 2 | long-form alias for canonical source | as source | dist_mets_bone | deterministic | AL |
| met_site_brain | 2 | long-form alias for canonical source | as source | dist_mets_brain_cns | deterministic | AL, G6 |
| met_site_soft_tissue | 2 | long-form alias for canonical source | as source | dist_mets_skin | deterministic | AL |
| met_site_visceral | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |
| sites_of_distant_metastasis_at_the_time_of_cancer_diagnosis_stage_iv_patients | X | Absent in cohort; always null, never sent to the model. | null | none | absent | X |

### P.5.7 Competing Risks module (C)

| Attribute | Type | Description | Allowed values / unit | Source mapping | Origin | Gates |
|---|---|---|---|---|---|---|
| tt_compete_first_dmets_mos | 2C | earliest non-brain distant met. Required when compete_first_dmets_event = 1. | float months | min(dx_to_dist_mets_<organ>_mos) across non-brain sites where dist_mets_<organ>==1 | computed | G2, C |
| compete_first_dmets_event | 2 | competing event indicator | 0/1 | any(dist_mets_<organ>==1) across non-brain sites | computed | G7 |

| Attribute | Type | Description | Allowed values / unit | Source mapping | Origin | Gates |
|---|---|---|---|---|---|---|
| `tt_brain_met_competing_mos` | 2 | multi-state time-to-first-event for Aim 2 competing-risks model | float months | `min(brain_time if event, other_time if event, OS_time)` | computed | G2 |
| `brain_met_competing_status` | 2 | competing-risks status for Aim 2 (cmprsk / FineGray) | 0 censor / 1 brain / 2 other | `argmin among brain/other/censor` | computed | G6 |

### P.5.8 Treatment module (M)

| Attribute | Type | Description | Allowed values / unit | Source mapping | Origin | Gates |
|---|---|---|---|---|---|---|
| `total_number_of_therapies_received_in_metastatic_disease_treatment` | 2 | long-form alias for canonical source | as source | `ca_n_regimens` | deterministic | AL |
| `chemo_overall` | 2 | any regimen with matching drug (GENIE BPC only) | 0/1 | regex on regimen_drugs: capecitabine\|cyclophosphamide\|paclitaxel\|docetaxel\|epirubici… **(truncated; full list pending)** | regex | R |
| `endocrine_overall` | 2 | any regimen with matching drug (GENIE BPC only) | 0/1 | regex on regimen_drugs: tamoxifen\|fulvestrant\|toremifene\|raloxifene\|anastrozole\|letr… **(truncated; full list pending)** | regex | G5 |
| `aromatase_inhibitor_overall` | 2 | any regimen with matching drug (GENIE BPC only) | 0/1 | regex on regimen_drugs: anastrozole\|letrozole\|exemestane… **(truncated; full list pending)** | regex | G5 |
| `tamoxifen_overall` | 2 | any regimen with matching drug (GENIE BPC only) | 0/1 | regex on regimen_drugs: tamoxifen\|toremifene\|raloxifene… **(truncated; full list pending)** | regex | G5 |
| `cdk4_6_inhibitor_overall` | 2 | any regimen with matching drug (GENIE BPC only) | 0/1 | regex on regimen_drugs: palbociclib\|ribociclib\|abemaciclib… **(truncated; full list pending)** | regex | R |
| `akt_inhibitor_overall` | 2 | any regimen with matching drug (GENIE BPC only) | 0/1 | regex on regimen_drugs: capivasertib\|ipatasertib\|miransertib\|afuresertib… **(truncated; full list pending)** | regex | R |
| `her2_targeted_overall` | 2 | any regimen with matching drug (GENIE BPC only) | 0/1 | regex on regimen_drugs: trastuzumab\|pertuzumab\|lapatinib\|neratinib\|tucatinib\|t-dm1\|a… **(truncated; full list pending)** | regex | R |
| `parp_inhibitor_overall` | 2 | any regimen with matching drug (GENIE BPC only) | 0/1 | regex on regimen_drugs: olaparib\|talazoparib\|niraparib\|rucaparib\|veliparib… **(truncated; full list pending)** | regex | R |

| Attribute | Type | Description | Allowed values / unit | Source mapping | Origin | Gates |
|---|---|---|---|---|---|---|
| `immuno_overall` | 2 | any regimen with matching drug (GENIE BPC only) | 0/1 | regex on regimen_drugs: pembrolizumab\|atezolizumab\|nivolumab\|ipilimumab\|durvalumab… **(truncated; full list pending)** | regex | R |

## P.6 Verification Gates

| ID | Gate | Rule |
|---|---|---|
| E | Enum | value is in the allowed set |
| R | Range and type | value has the declared type and unit; numeric ranges per attribute |
| X | Absent | type-X attributes are `null` with `origin = absent` |
| AL | Alias agreement | aliases of one source column hold identical values |
| AN | Source anchor | model-filled values quote the source exactly at their offsets |
| O | Ontology | OncoTree and ICD-O-3 codes exist in the frozen code lists |
| C | Conditional presence | 1C and 2C attributes present exactly when their condition holds |
| G1 | Stage vs flag | `stage_diag_group = Stage IV` iff `stage_iv_bin = 1` |
| G2 | Within follow-up | time-to-event values <= `overall_survival_months` |
| G3 | Age consistency | birth year plus age at diagnosis not in the future; `age_cat` matches age |
| G4 | Subtype vs HER2 | Triple Negative and HR+/HER2- need HER2 = 0; HER2+ subtypes need 1 |
| G5 | Drug vs class | tamoxifen or aromatase inhibitor = 1 requires `endocrine_overall = 1` |
| G6 | Brain status vs site | `brain_met_competing_status = 1` requires `distant_mets_brain = 1` |
| G7 | Competing event vs sites | `compete_first_dmets_event = 1` iff any non-brain site = 1 |

Gates report `skipped` when an input is `null`; missingness never causes a pass or fail.

## P.7 Output Mapping

| Module | `x_study_features` section |
|---|---|
| General Study, Patient, Clinical | `clinical` |

| Module | `x_study_features` section |
|---|---|
| Pathology, Sequencing | `pathology` |
| Follow-up, Competing Risks | `follow_up` |
| Treatment | `treatment` |

## P.8 Open Issues

| # | Issue | Resolution needed before |
|---|---|---|
| 1 | `patients_age_of_death_in_days` maps to `age_death_yrs` (years). | protocol 1.0 |
| 2 | `age_at_metastatic_diagnosis` maps to `dx_to_dmets_mos`, an interval from diagnosis, not an age. | protocol 1.0 |
| 3 | `age_at_first_distant_metastasis_in_days` maps to `dx_to_dmets_days`, an interval from diagnosis, not an age. | protocol 1.0 |
| 4 | Drug-class regexes in the dictionary are truncated (`...`); full lists must be frozen in `drug_lists.yaml`. | protocol 1.0 |
| 5 | Allowed ranges for continuous attributes (ages, months, scores) need clinical sign-off. | protocol 1.0 |
| 6 | GENIE BPC is controlled access; confirm the data use agreement covers this processing. | first real run |